CO2 AT3

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Q1. Data Cleaning and Normalization in Healthcare Analytics A.

Missing Value Treatment

i. Missing Values

Column	Missing Record
Age	P2
BMI	P3
Annual_Visits	P5

ii. Mean and Median Imputation

Age

Available values: 25, 42, 110, 36, 50, 29, 45

Mean = $\frac{25+42+110+36+50+29+45}{7} = \frac{337}{7} = 48.14$

Median

Sorted:

25, 29, 36, **42**, 45, 50, 110

Median = **42**

Imputed Age

- Mean → **48.14**
- Median → **42**

BMI

Available

22.5,24.8,45.2,29.5,31.2,21.8,60

Mean $= \frac{235}{7} = 33.57$

Median

Sorted

21.8,22.5,24.8,**29.5**,31.2,45.2,60

Median = **29.5**

Annual Visits

Available

3,5,7,50,12,4,40

Mean $= \frac{121}{7} = 17.29$

Median

Sorted

3,4,5,**7**,12,40,50

Median = **7**

iii. Comparison

Attribute	**Mean**	**Median**	**Better**
Age	48.14	42	Median
BMI	33.57	29.5	Median
Annual Visits	17.29	7	Median

Justification: Median is preferred because outliers (Age =110, Visits =50) affect the mean.

B. Outlier Detection

Using Z-score: $Z = \frac{x-\mu}{\sigma}$

Potential Outliers

- Age =110
- Annual_Visits =50

These records have unusually high values.

Treatment

Use capping instead of deletion because medical data are valuable.

C. Categorical Standardization

Inconsistent Values

M, Male, male

F, Female, female, FEMALE

Standardized

Male

Female

Importance

- Removes duplicate categories.
- Improves model accuracy.
- Prevents encoding errors.
- Reduces preprocessing complexity.

D. Min-Max Normalization (BMI)

Minimum =21.8

Maximum =60

Formula

$$x' = \frac{x - 21.8}{60 - 21.8}$$

BMI	Normalized
22.5	0.018

24.8	0.079
29.5	0.201
31.2	0.246
45.2	0.613
BMI	**Normalized**
60	1.000

Missing BMI is imputed before normalization.

Effect

- Brings values between 0 and 1.
- Faster model convergence.
- Prevents large-scale features dominating.

Q2. Covariance and Feature Selection

A. Covariance

Income

35,50,70,100,42,65,130,85

Loan

100,150,200,300,120,180,350,250

Mean Income =72.125

Mean Loan =206.25

Covariance : $Cov(X, Y) = \Sigma \frac{(x-\bar{x})(y-\bar{y})}{n-1}$

Result : $Cov \approx 2915$

Positive covariance.

Interpretation

Higher income customers generally receive larger loans.

B. Highly Correlated Features

Possible correlated pairs

- Income ↔ Loan Amount
- Income ↔ Savings
- Age ↔ Income
- Savings ↔ Loan Amount

Redundant information exists because higher-income people usually have higher savings and loans.

Features that could be removed

- Savings
- Monthly EMI
- Age

C. Feature Selection

Keep

- Income
- Credit Score
- Loan Amount
- Risk

Reason

These directly influence customer risk.

Feature selection

- Reduces computation
- Removes redundancy
- Improves accuracy
- Reduces overfitting

D. Z-score Standardization

Formula : $Z = \frac{x-\mu}{\sigma}$

Apply to

- Age
- Income
- Loan Amount
- Credit Score
- EMI
- Savings **Why Scaling?**

Income is measured in ₹ thousands whereas Credit Score ranges around 650–820.

Without scaling, Income dominates distance-based algorithms.

Q3. Discretization of Sales Data

A. Equal Width Binning

Minimum =500

Maximum =6500

Range

6000

Bin Width $= \frac{6000}{4} = 1500$

Intervals

Bin	Interval
1	500–2000
2	2000–3500
3	3500–5000
4	5000–6500

Assignments

Customer	Bin
C1-C4	Bin1
C5-C7	Bin2

C8	Bin3
C9-C10	Bin4

B. Equal Frequency Binning 10

records

Approximately 3 records/bin

Bin	Customers
1	C1,C2,C3
2	C4,C5,C6
3	C7,C8
4	C9,C10

Comparison

Equal Width

- Same interval size
- Unequal records

Equal Frequency

- Equal records
- Unequal interval size

C. Concept Hierarchy

Monthly Spend

↓

Low

↓

Medium

↓

High

↓

Very High

Example

Low →500–1500

Medium →1501–3000

High →3001–5000

Very High →Above5000

D. Interpretation

Advantages

- Easier interpretation
- Better visualization
- Simpler rule generation
- Useful in decision trees

Disadvantage

Information loss due to grouping.

Q4. Manufacturing Data Transformation

A. Min-Max Normalization Minimum

=55

Maximum =120

Formula: $x' = \frac{x-55}{65}$

Temperature	**Normalized**
55	0.000
60	0.077
62	0.108
65	0.154

70	0.231
75	0.308
80	0.385
120	1.000

Maximum normalized value

Machine **M8**

B. Z-score

Mean $= \frac{587}{8} = 73.38$

Standard deviation

≈19.64

Formula: $Z = \frac{x - 73.38}{19.64}$

Calculate Z-score for each value.

C. Outlier 120°C

Z≈2.37

Usually

|Z|>2 indicates an extreme value.

Effect

Raises the mean.

Compresses normalized values in Min-Max scaling.

Treatment

Use capping or Winsorization.

D. Comparison

Min-Max	**Z-score**

Range 0–1	Mean 0 SD 1
Sensitive to outliers	Less sensitive
Good for Neural Networks	Good for statistical analysis

Q5. Student Analytics

A. Missing Values

Column	**Missing**
Internal Mark	S2
Column	**Missing**
Attendance	S3
Study Hours	S5
Assignment Score	S6

Mean Imputation

Study Hours

Mean = $\frac{2+3+4+5+7+8+12}{7} = 5.86$

Median

Sorted

2,3,4,5,7,8,12

Median =5

Better Method

Median is preferred because Study_Hours has an outlier (12).

B. Data Transformation

Internal Marks Min

=55

Max =98 Formula:

$$x' = \frac{x-55}{43}$$

Apply Min-Max normalization.

Z- Score

First fill missing value using mean.

Then apply: $Z = \frac{x-\mu}{\sigma}$

Comparison

- Min-Max → values between 0 and 1.
- Z-score → values centered around 0.

C. Discretization

Suggested thresholds

Category	Study Hours
Low	0–4
Medium	5–8
High	Above 8

Assignments

Student	Category
S1	Low
S2	Low
S3	Low
S4	Medium
S5	Medium
S6	Medium

S7	Medium
S8	High

D. Interpretation

Data preprocessing improves model performance by:

- Handling missing values.
- Removing outliers.
- Standardizing categorical values.
- Scaling numerical attributes.
- Reducing noise.
- Improving prediction accuracy.
- Making models easier to interpret.